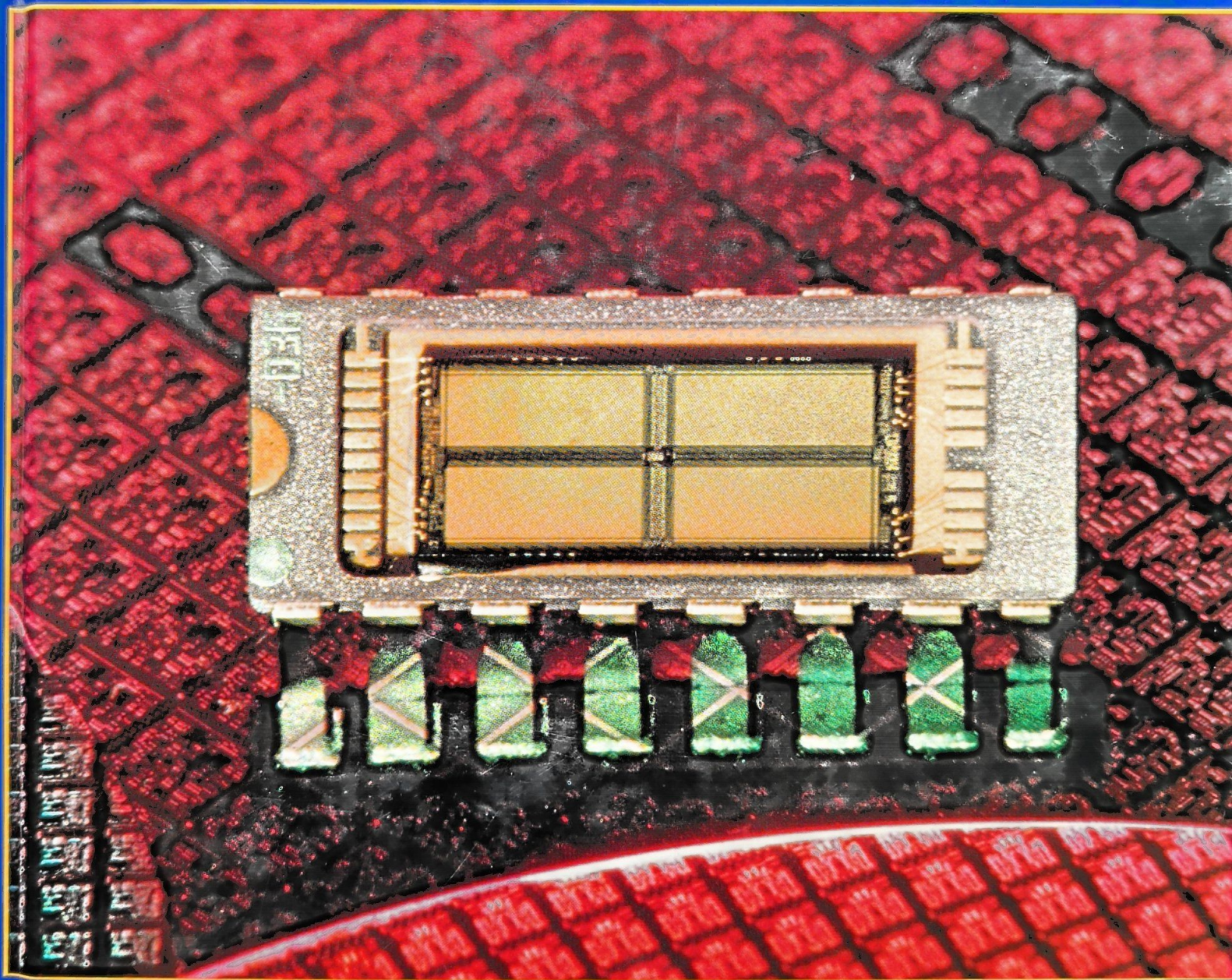


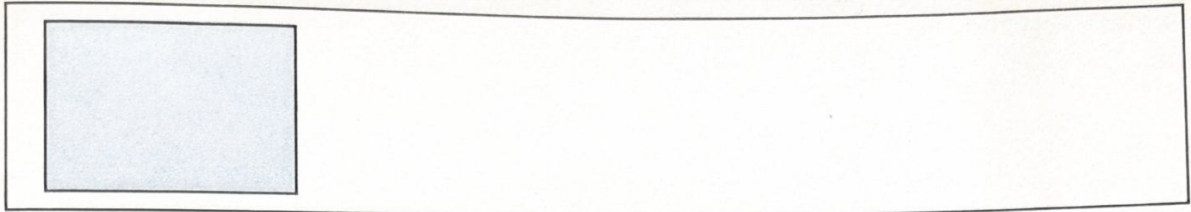
EXHIBIT 33

MICROELECTRONIC CIRCUITS

THIRD EDITION



SEDRA / SMITH



Microelectronic Circuits

**THIRD
EDITION**

Microelectronic Circuits

Adel S. Sedra
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ABOUT THE COVER

In the background is a processed wafer of silicon on which a large number of identical circuit chips are fabricated. When cut from such a wafer, a chip can be housed in a package such as the one shown here with its cover removed. This single chip is a segmented memory array with a very large number of active devices in a variety of intricate circuit structures. The growing need for efficient and reliable low-cost components such as these underlies the importance of the various circuit design and circuit analysis techniques discussed in this text.

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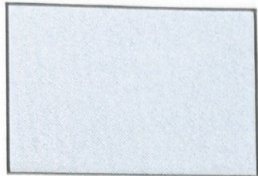
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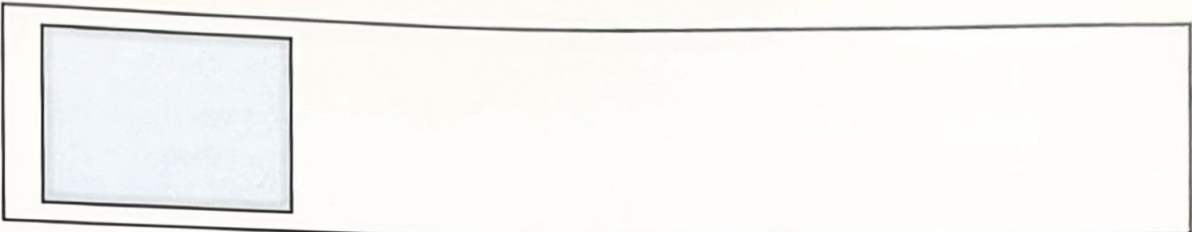
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INTRODUCTION

In this chapter we study the other major type of transistor, the field-effect transistor (FET). As in the case of the BJT (Chapter 4), the voltage between two terminals of the FET controls the current flow in the third terminal. Correspondingly, the FET can be used both as an amplifier and as a switch.

The field-effect transistor derives its name from the essence of its physical operation. Specifically, it will be shown that the current-control mechanism is based on an electric field established by the voltage applied to the control terminal. It will also be shown that current is conducted by only one type of carrier (electrons or holes) depending on the type of FET (n channel or p channel), which gives the FET another name, the **unipolar transistor**.

Although the basic concept of the FET has been known since the 1930s, the device became a practical reality only in the 1960s. Since the late 1970s, a particular kind of FET, the **metal-oxide semiconductor field-effect transistor** (MOSFET), has been ex-

tremely popular. Compared to BJTs, MOS transistors can be made quite small (that is, occupying a small silicon area on the IC chip), and their manufacturing process is relatively simple (see Appendix A). Furthermore, digital logic and memory functions can be implemented with circuits that exclusively use MOSFETs (that is, no resistors or diodes are needed). For these reasons, most very-large-scale integrated (VLSI) circuits are made at the present time using MOS technology. Examples include microprocessor and memory chips. MOS technology has also been applied extensively in the design of analog integrated circuits.

Although the FET family of devices has many different types, and we shall discuss a few, most of the chapter is devoted to the enhancement-type MOSFET, which is by far the most significant field-effect transistor. Its significance is on par with that of the bipolar junction transistor, with each having its own areas of application.

Field-effect transistors are available in discrete form, and we shall study their application in discrete circuit design. Their most important use, however, is in the design of integrated circuits. The study of MOS integrated circuits begins in this chapter and is continued in subsequent chapters.

5.1 STRUCTURE AND PHYSICAL OPERATION OF THE ENHANCEMENT-TYPE MOSFET

The enhancement-type MOSFET is the most widely used field-effect transistor. In this section we shall study its structure and physical operation. This will lead to the current-voltage characteristics of the device, studied in the next section.

Device Structure

Figure 5.1 shows the physical structure of the n -channel enhancement-type MOSFET. The meaning of the names “enhancement” and “ n -channel” will become apparent shortly. The transistor is fabricated on a p -type substrate, which is a single-crystal silicon wafer that provides physical support for the device (and for the entire circuit in the case of an integrated circuit). Two heavily doped n -type regions, indicated in the figure as the n^+ **source**¹ and the n^+ **drain** regions, are created in the substrate. A thin (about $0.1\ \mu\text{m}$) layer² of silicon dioxide (SiO_2), which is an excellent electrical insulator, is grown on the surface of the substrate, covering the area between the source and drain regions. Metal is deposited on top of the oxide layer to form the **gate electrode** of the device. Metal contacts are also made to the source region, the drain region, and the substrate, also known as the **body**. Thus, four terminals are brought out: the gate terminal (G), the source terminal (S), the drain terminal (D), and the substrate or body terminal (B).

At this point it should be clear that the name of the device (metal-oxide-semiconductor FET) is derived from its physical structure. The name, however, has become a general one and is used also for FETs that do not use metal for the gate electrode. In fact, most

¹The notation n^+ indicates heavily doped n -type silicon. Conversely, n^- is used to denote lightly doped n -type silicon. Similar notation applies for p -type silicon.

²A micrometer (μm), also called micron, is $10^{-6}\ \text{m}$.

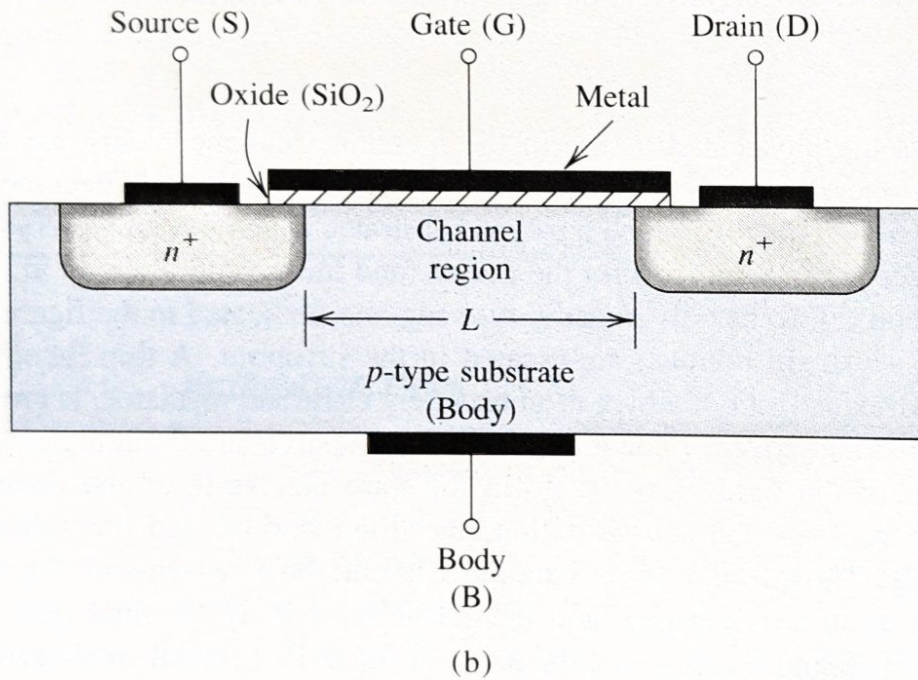
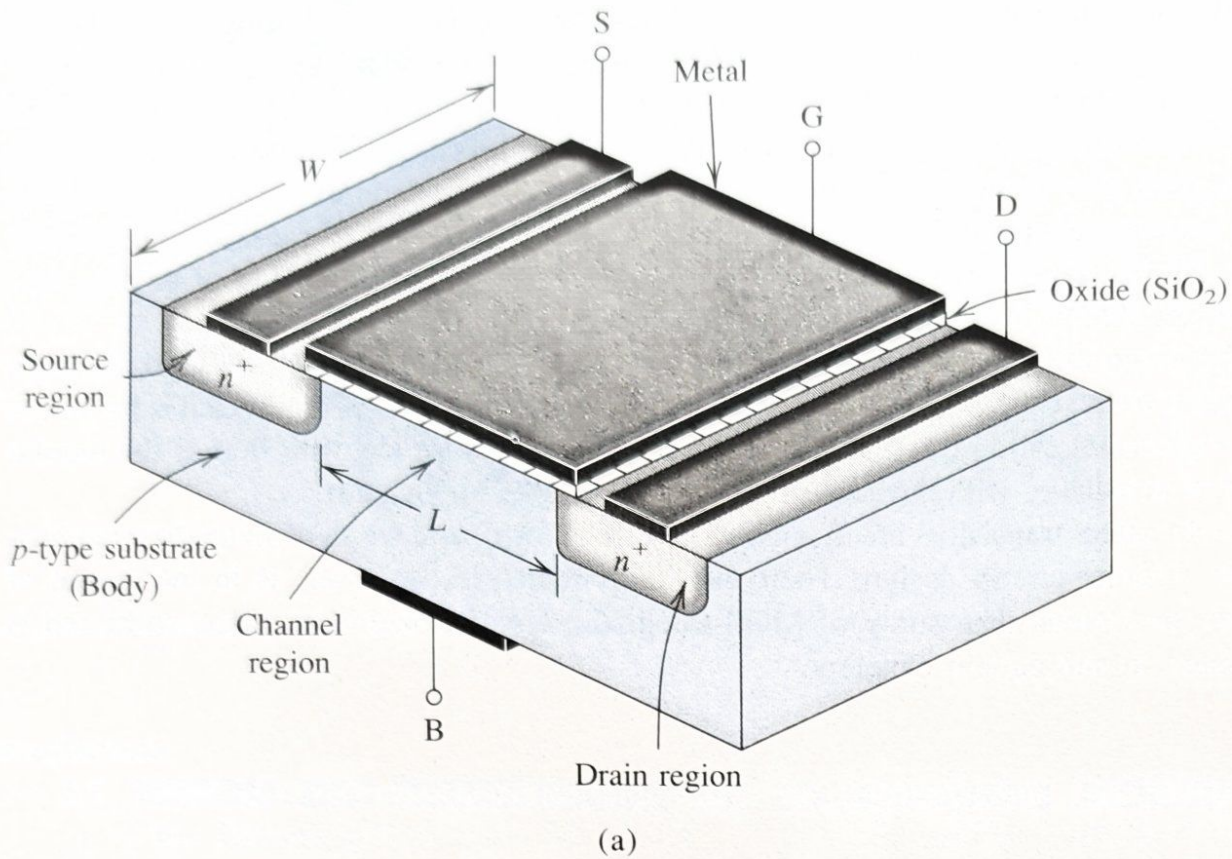


Fig. 5.1 Physical structure of enhancement-type NMOS transistor: **(a)** perspective view; **(b)** cross section. Typically $L = 1$ to $10\ \mu\text{m}$, $W = 2$ to $500\ \mu\text{m}$, and the thickness of the oxide layer is of the order of $0.1\ \mu\text{m}$.

modern MOSFETs are fabricated using a process known as silicon-gate technology, in which a certain type of silicon, called polysilicon, is used to form the gate electrode (see Appendix A). Our description of MOSFET operation and characteristics apply irrespective of the type of gate electrode.

Another name for the MOSFET is the **insulated-gate FET** or **IGFET**. This name also arises from the physical structure of the device, emphasizing the fact that the gate electrode is electrically insulated from the device body (by the oxide layer). It is this insulation that causes the current in the gate terminal to be extremely small (of the order of 10^{-15} A).

Observe that the substrate forms *pn* junctions with the source and drain regions. In normal operation these *pn* junctions are kept reverse-biased at all times. Since the drain will be at a positive voltage relative to the source, the two *pn* junctions can be effectively cut off by simply connecting the substrate terminal to the source terminal. We shall assume this to be the case in the following description of MOSFET operation. Thus, here, the substrate will be considered as having no effect on device operation, and the MOSFET will be treated as a three-terminal device, with the terminals being the gate (G), the source (S), and the drain (D). It will be shown that a voltage applied to the gate controls current flow between source and drain. This current will flow in the longitudinal direction from drain to source in the region labeled “channel region.” Note that this region has a length L and a width W , two important parameters of the MOSFET. Typically, L is in the range 1 to 10 μm , and W is in the range 2 to 500 μm . Finally note that, unlike the BJT, the MOSFET is normally constructed as a symmetrical device. Thus its source and drain can be interchanged with no change in device characteristics.

Operation with No Gate Voltage

With no bias voltage applied to the gate, two back-to-back diodes exist in series between drain and source. One diode is formed by the *pn* junction between the n^+ drain region and the *p*-type substrate, and the other diode is formed by the *pn* junction between the *p*-type substrate and the n^+ source region. These back-to-back diodes prevent current conduction from drain to source when a voltage v_{DS} is applied. In fact, the path between drain and source has a very high resistance (of the order of 10^{12} Ω).

Creating a Channel for Current Flow

Consider next the situation depicted in Fig. 5.2. Here we have grounded the source and the drain and applied a positive voltage to the gate. Since the source is grounded, the gate voltage appears in effect between gate and source and thus is denoted v_{GS} . The positive voltage on the gate causes in the first instance the free holes (which are positively charged) to be repelled from the region of the substrate under the gate (the channel region). These holes are pushed downward into the substrate, leaving behind a carrier-depletion region. The depletion region is populated by the bound negative charge associated with the acceptor atoms. These charges are “uncovered” because the neutralizing holes have been pushed downward into the substrate.

As well, the positive gate voltage attracts electrons from the n^+ source and drain regions (where they are in abundance) into the channel region. When a sufficient number of electrons accumulate near the surface of the substrate under the gate, an *n* region is in

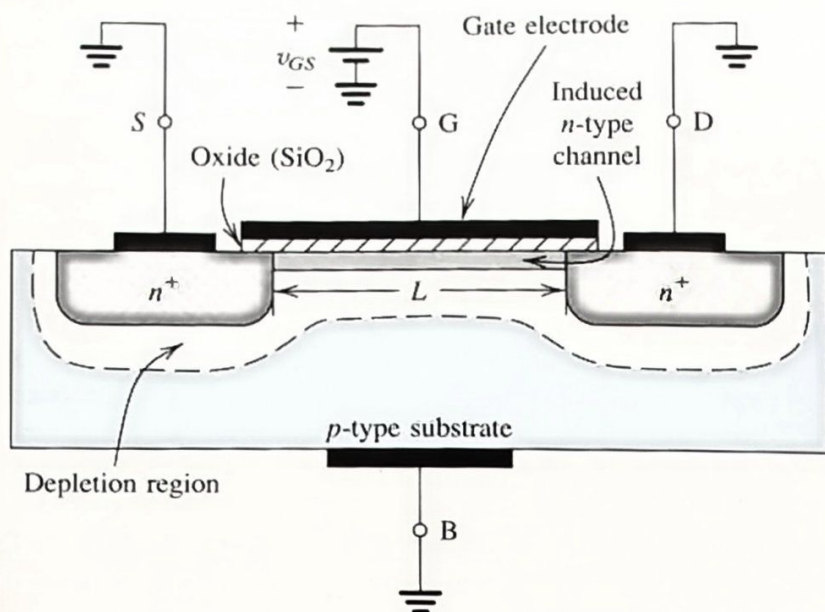


Fig. 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

effect created, connecting the source and drain regions, as indicated in Fig. 5.2. Now if a voltage is applied between drain and source, current flows through this induced n region, carried by the mobile electrons. The induced n region thus forms a **channel** for current flow from drain to source and is aptly called so. Correspondingly, the MOSFET of Fig. 5.2 is called an **n -channel MOSFET** or, alternatively, an **NMOS transistor**. Note that an n -channel MOSFET is formed in a p -type substrate: The channel is created by inverting the substrate surface from p type to n type. Hence the induced channel is called an **inversion layer**.

The value of v_{GS} at which a sufficient number of mobile electrons accumulate in the channel region to form a conducting channel is called the **threshold voltage** and is denoted V_t .³ Obviously, V_t for an n -channel FET is positive. The value of V_t is controlled during device fabrication and typically lies in the range 1 to 3 V.

The gate and body of the MOSFET form a parallel-plate capacitor with the oxide layer acting as the capacitor dielectric. The positive gate voltage causes positive charge to accumulate on the top plate of the capacitor (the gate electrode). The corresponding negative charge of the bottom plate is formed by the electrons in the induced channel. An electric field thus develops in the vertical direction. It is this field that controls the amount of charge in the channel, and thus it determines the channel conductivity and, in turn, the current that will flow through the channel when a voltage v_{DS} is applied.

Applying a Small v_{DS}

Having induced a channel, we now apply a positive voltage v_{DS} between drain and source, as shown in Fig. 5.3. We first consider the case where v_{DS} is small (say, 0.1 or 0.2 V).

³Many texts use V_T to denote the threshold voltage. We use V_t to avoid confusion with the thermal voltage V_T .

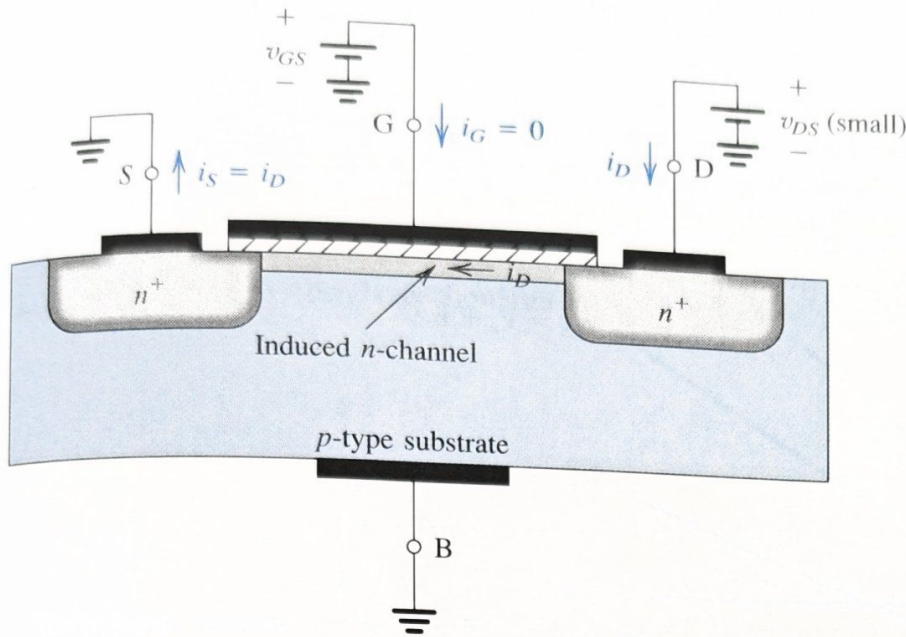


Fig. 5.3 An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied. The device acts as a conductance whose value is determined by v_{GS} . Specifically, the channel conductance is proportional to $v_{GS} - V_t$ and thus i_D is proportional to $(v_{GS} - V_t)v_{DS}$. Note that the depletion region is not shown (for simplicity).

The voltage v_{DS} causes a current i_D to flow through the induced n channel. Current is carried by free electrons traveling from source to drain (hence the names source and drain). The magnitude of i_D depends on the density of electrons in the channel, which in turn depends on the magnitude of v_{GS} . Specifically, for $v_{GS} = V_t$ the channel is just induced and the current conducted is still negligibly small. As v_{GS} exceeds V_t , more electrons are attracted into the channel. We may visualize the increase in charge carriers in the channel as an increase in the channel depth. The result is a channel of increased conductance or equivalently reduced resistance. In fact, the conductance of the channel is proportional to the **excess gate voltage** ($v_{GS} - V_t$). It follows that the current i_D will be proportional to $v_{GS} - V_t$ and, of course, to the voltage v_{DS} that causes i_D to flow.

Figure 5.4 shows a sketch of i_D versus v_{DS} for various values of v_{GS} . We observe that the MOSFET is operating as a linear resistance whose value is controlled by v_{GS} . The resistance is infinite for $v_{GS} \leq V_t$, and its value decreases as v_{GS} exceeds V_t .

The description above indicates that for the MOSFET to conduct, a channel has to be induced. Then, increasing v_{GS} above the threshold voltage V_t enhances the channel, hence the names **enhancement-mode operation** and **enhancement-type MOSFET**. Finally, we note that the current that leaves the source terminal (i_S) is equal to the current that enters the drain terminal (i_D), and the gate current $i_G = 0$.

Operation as v_{DS} is Increased

We next consider the situation as v_{DS} is increased. For this purpose let v_{GS} be held constant at a value greater than V_t . Refer to Fig. 5.5, and note that v_{DS} appears as a voltage drop across the length of the channel. That is, as we travel along the channel from source

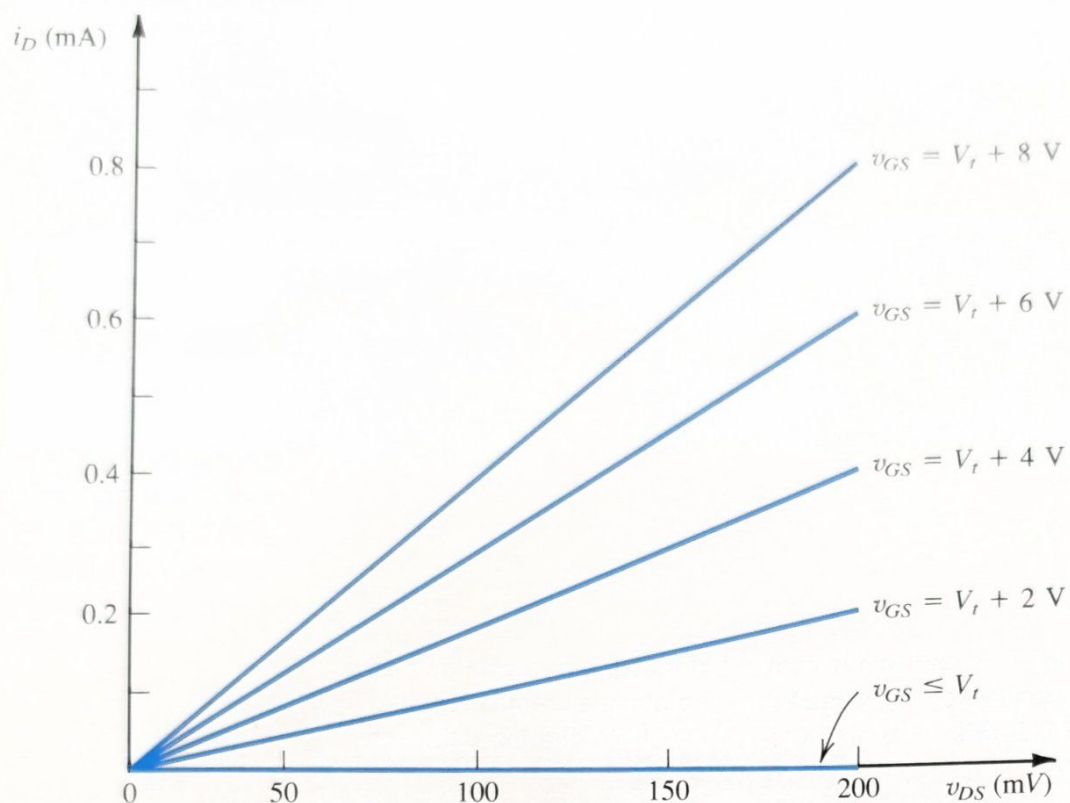


Fig. 5.4 The i_D - v_{DS} characteristics of the MOSFET in Fig. 5.3. Note that v_{DS} is small. The device operates as a linear resistor whose value is controlled by v_{GS} .

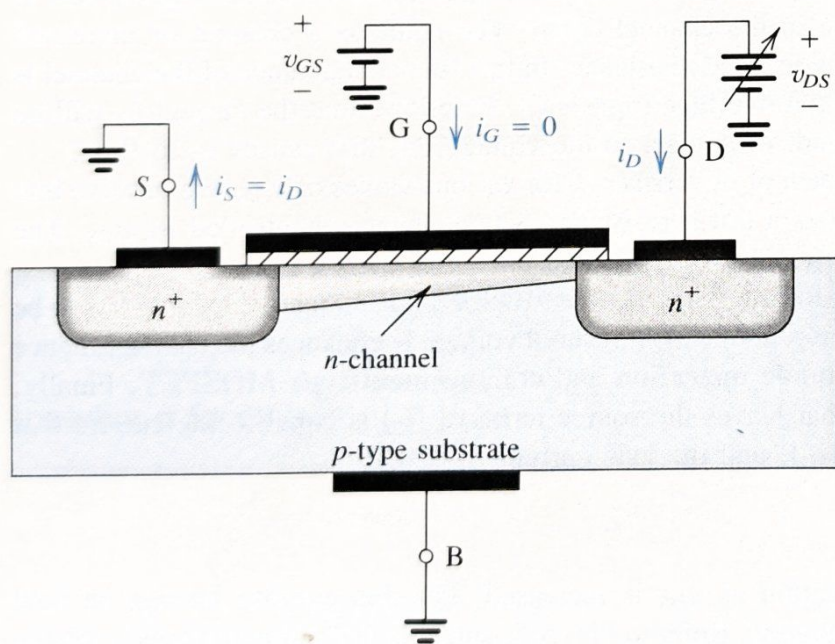


Fig. 5.5 Operation of the enhancement NMOS' transistor as v_{DS} is increased. The induced channel acquires a tapered shape and its resistance increases as v_{DS} is increased. Here v_{GS} is kept constant at a value $> V_t$.

to drain, the voltage (measured relative to the source) increases from 0 to v_{DS} . Thus the voltage between the gate and points along the channel decreases from v_{GS} at the source end to $v_{GS} - v_{DS}$ at the drain end. Since the channel depth depends on this voltage, we find that the channel is no longer of uniform depth; rather, the channel will take the tapered form shown in Fig. 5.5, being deepest at the source end and shallowest at the drain end. As v_{DS} is increased, the channel becomes more tapered and its resistance increases correspondingly. Thus the i_D - v_{DS} curve does not continue as a straight line but bends as shown in Fig. 5.6. Eventually, when v_{DS} is increased to the value that reduces

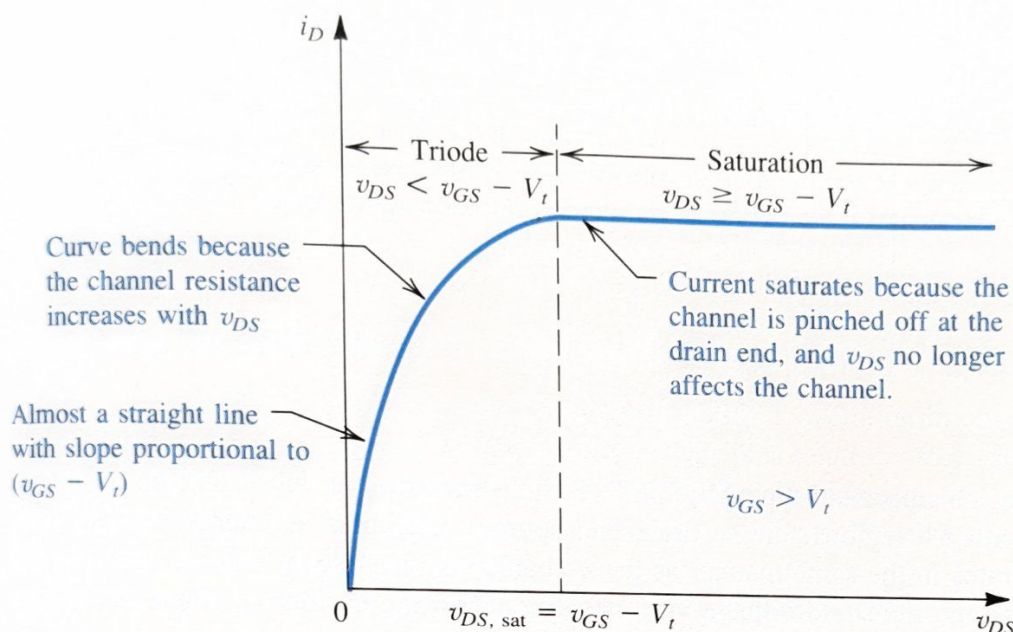


Fig. 5.6 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} > V_t$.

the voltage between gate and channel at the drain end to V_t —that is, $v_{GS} - v_{DS} = V_t$ or $v_{DS} = v_{GS} - V_t$ —the channel depth at the drain end decreases to almost zero, and the channel is said to be **pinched off**. Increasing v_{DS} beyond this value has little effect (theoretically, no effect) on the channel shape, and the current through the channel remains constant at the value reached for $v_{DS} = v_{GS} - V_t$. The drain current thus **saturates** at this value, and the MOSFET is said to have entered the **saturation region**⁴ of operation. The voltage v_{DS} at which saturation occurs is denoted $v_{DS,sat}$,

$$v_{DS,sat} = v_{GS} - V_t \quad (5.1)$$

Obviously, for every value of $v_{GS} \geq V_t$, there is a corresponding value of $v_{DS,sat}$. The device operates in the saturation region if $v_{DS} \geq v_{DS,sat}$. The region of the i_D - v_{DS} charac-

⁴ Saturation in a MOSFET means a very different thing than in a BJT. This is a rather unfortunate situation but cannot be changed at this time: Virtually the entire electronics literature uses this terminology!

teristic obtained for $v_{DS} < v_{DS,sat}$ is called the **triode region**, a carryover from the days of vacuum-tube devices whose operation an FET resembles.

To help further in visualizing the effect of v_{DS} , we show in Fig. 5.7 sketches of the channel as v_{DS} is increased while v_{GS} is kept constant. Theoretically, any increase in v_{DS} above $v_{DS,sat}$ (which is equal to $v_{GS} - V_t$) has no effect on the channel shape and simply appears across the depletion region surrounding the channel and the n^+ drain region.

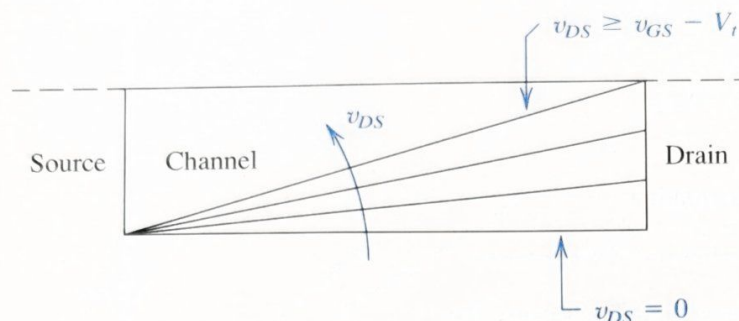


Fig. 5.7 Increasing v_{DS} causes the channel to acquire a tapered shape. Eventually, as v_{DS} reaches $v_{GS} - V_t$, the channel is pinched off at the drain end. Increasing v_{DS} above $v_{GS} - V_t$ has little effect (theoretically, no effect) on the channel's shape.

The p-channel MOSFET

A p-channel enhancement-type MOSFET (PMOS transistor) is fabricated on an n -type substrate with p^+ regions for the drain and source, and holes as charge carriers. The device operates in the same manner as the n -channel device except that v_{GS} and v_{DS} are negative and the threshold voltage V_t is negative. Also the current i_D enters the source terminal and leaves through the drain terminal.

PMOS technology was originally the dominant one. However, because NMOS devices can be made smaller and thus operate faster, and because NMOS requires lower supply voltages than PMOS, NMOS technology has virtually replaced PMOS. Nevertheless, it is important to be familiar with the PMOS transistor for two reasons: PMOS devices are still available for discrete-circuit design, and more importantly, both PMOS and NMOS transistors are utilized in **complementary MOS** or **CMOS** circuits.

Complementary MOS or CMOS

As the name implies, complementary MOS technology employs MOS transistors of both polarities. Although CMOS circuits are somewhat more difficult to fabricate than NMOS, the availability of complementary devices makes possible many powerful circuit-design possibilities. Indeed, at the present time CMOS is the most useful of all the integrated-circuit MOS technologies. This statement applies to both analog and digital circuits. Throughout this book we will study many CMOS circuit techniques.

Figure 5.8 shows a cross section of a CMOS chip illustrating how the PMOS and NMOS transistors are fabricated. Observe that while the PMOS transistor is implemented directly in the n -type substrate, the NMOS transistor is fabricated in a specially created p region, known as a p well. The two devices are isolated from each other by a thick region of oxide that functions as an insulator.

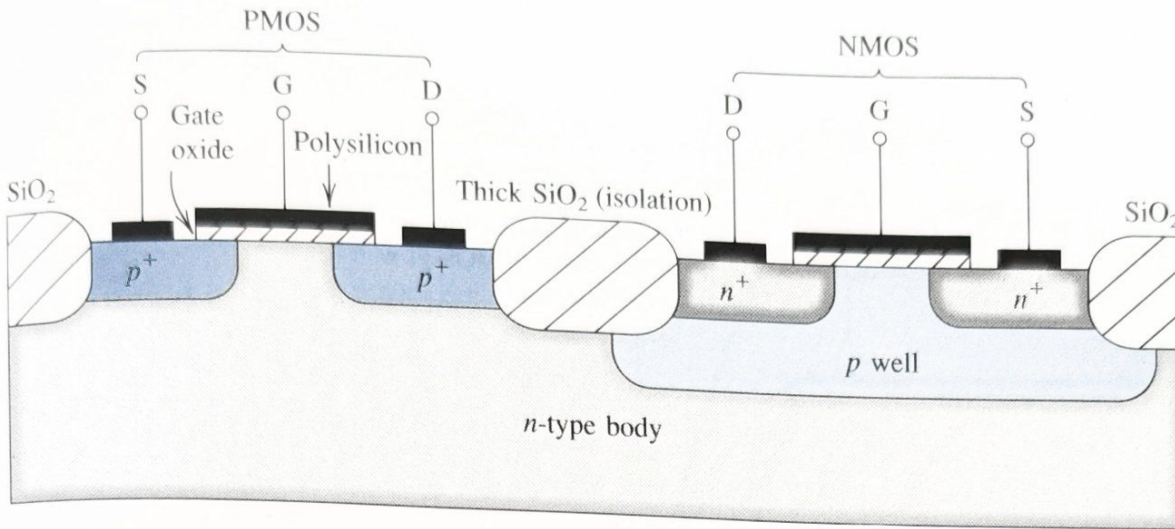


Fig. 5.8 Cross section of a CMOS integrated circuit. Note that the NMOS transistor is formed in a separate p-type region, known as a p well. Another arrangement is also possible in which a p-type body is used and the p device is formed in an n well.

Operating the MOS Transistor in the Subthreshold Region

The above description of MOSFET operation implies that for $v_{GS} < V_t$, no current flows and the device is cut off. This is not entirely true for it has been found that for values of v_{GS} smaller but close to V_t , a small drain current flows. In this **subthreshold region** of operation the drain current is exponentially related to v_{GS} , much like the i_C - v_{BE} relationship of a BJT.

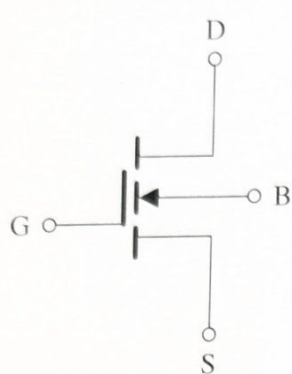
Although in most applications the MOS transistor is operated with $v_{GS} > V_t$, there are special, but a growing number of, applications that make use of subthreshold operation. In this book we will not consider subthreshold operation any further and refer the reader to the references listed at the end of the chapter [see in particular Mead (1988) and Tsividis (1987)].

5.2 CURRENT-VOLTAGE CHARACTERISTICS OF THE ENHANCEMENT MOSFET

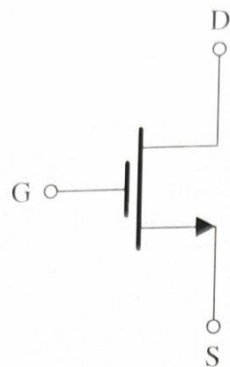
Building on the physical foundation established in the previous section for the operation of the enhancement MOS transistor, we present in this section its complete current-voltage characteristics. These characteristics can be measured at dc or at low frequencies and thus are called static characteristics. The dynamic effects that limit the operation of the MOSFET at high frequencies and high switching speeds will be discussed in Chapter 7 for amplifier circuits, and in Chapter 13 for digital circuits.

Circuit Symbol

Figure 5.9(a) shows the circuit symbol for the n-channel enhancement-type MOSFET. The symbol is very descriptive: The vertical solid line denotes the gate electrode; the vertical broken line denotes the channel—the line is broken to indicate that the device is of the enhancement type whose channel does not exist without the application of an



(a)



(b)

Fig. 5.9 (a) Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Simplified circuit symbol for the enhancement *n*-channel MOSFET with the substrate (B) connected to the source (S).

appropriate gate voltage; and the spacing between the gate line and the channel line represents the fact that the gate electrode is insulated from the body of the device. The polarity of the *pn* junction between the *p*-type substrate and the *n* channel is indicated by the arrowhead on the line representing the substrate (body). This arrowhead also indicates the polarity of the transistor; namely, that it is an *n*-channel device.

Although the MOSFET is a symmetrical device, it is useful in circuit applications to designate one terminal as the source and the other as the drain (without having to write S and D beside the terminals). This goal is achieved in the circuit symbol by drawing the gate terminal line closer to the source than to the drain.⁵

Although the symbol of Fig. 5.9(a) is descriptive it is rather complex and makes drawing a large circuit an awkward task. A simplified circuit symbol that applies for the usual case where the substrate is connected to the source is shown in Fig. 5.9(b). In this symbol, the arrowhead on the source terminal points in the normal direction of current flow and accomplishes two purposes: It distinguishes the source from the drain, and it indicates the polarity of the device (i.e., *n* channel).

The i_D - v_{DS} Characteristics

Figure 5.10(a) shows an *n*-channel enhancement-type MOSFET with voltages v_{GS} and v_{DS} applied and with the normal directions of current flow indicated. This conceptual circuit can be used to measure the i_D - v_{DS} characteristics, which are a family of curves, each measured at a constant v_{GS} . From the study of physical operation in the previous section, we expect each of the i_D - v_{DS} curves to have the shape shown in Fig. 5.6. This indeed is the case, as is evident from Fig. 5.10(b) which shows a typical set of i_D - v_{DS} characteristics. A thorough understanding of the MOSFET terminal characteristics is essential for the reader who intends to design MOS circuits.

The characteristic curves in Fig. 5.10(b) indicate that there are three distinct regions of operation: the cutoff region, the triode region, and the saturation region. The saturation region is used if the FET is to operate as an amplifier. For operation as a switch, the cutoff

⁵ In practice, it is the polarity of the voltage impressed across the device that determines source and drain; the drain is always positive relative to the source in an *n*-channel FET.

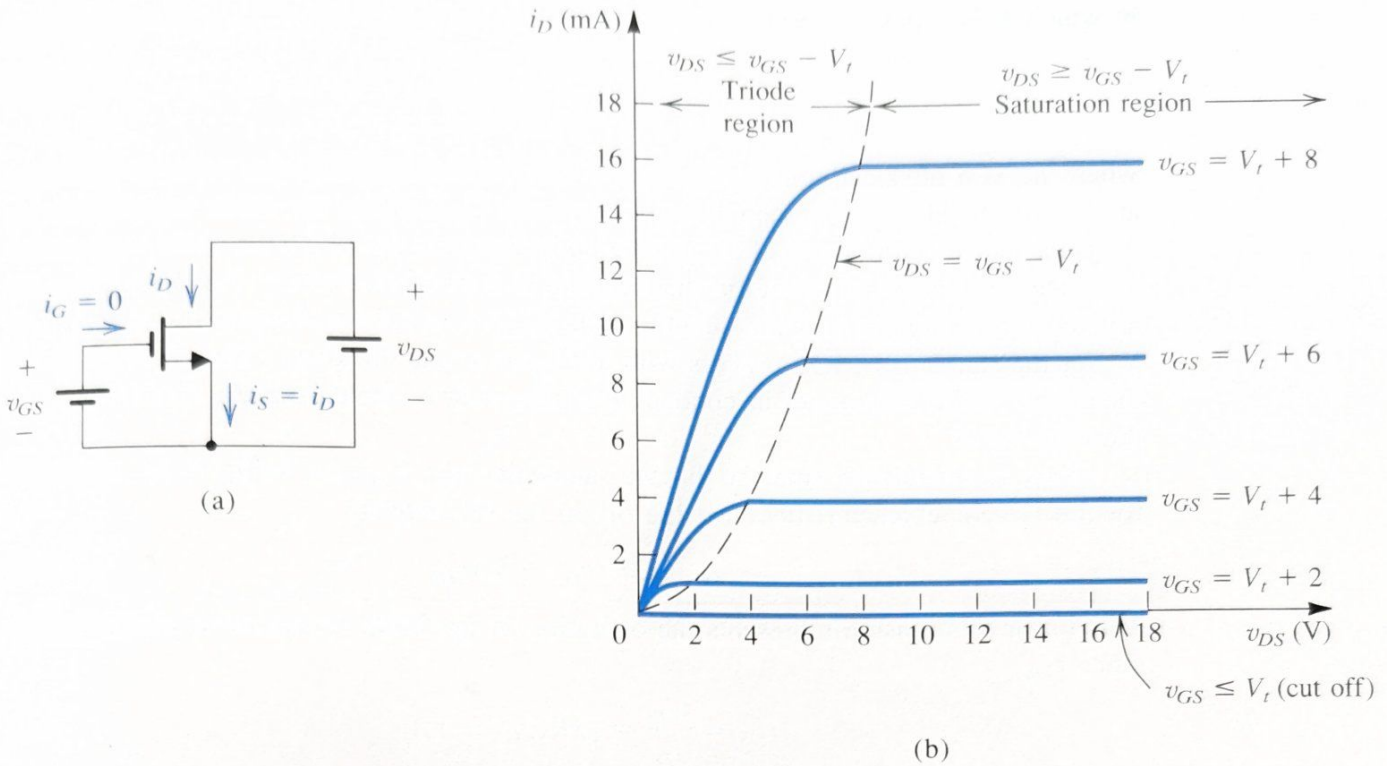


Fig. 5.10 (a) An n -channel enhancement-type MOSFET with v_{GS} and v_{DS} applied and with the normal directions of current flow indicated. **(b)** The i_D - v_{DS} characteristics for a device with $V_t = 2$ V and $K = 0.25$ mA/V².

and triode regions are utilized. The device is cut off when $v_{GS} < V_t$. To operate the MOSFET in the triode region we must first induce a channel,

$$v_{GS} \geq V_t \quad (\text{Induced channel}) \quad (5.2)$$

and then keep v_{DS} small enough so that the channel remains continuous. This is achieved by ensuring that the gate-to-drain voltage is

$$v_{GD} > V_t \quad (\text{Continuous channel}) \quad (5.3)$$

This condition can be stated explicitly in terms of v_{DS} by writing $v_{GD} = v_{GS} + v_{SD} = v_{GS} - v_{DS}$; thus

$$v_{GS} - v_{DS} > V_t$$

which can be rearranged to yield

$$v_{DS} < v_{GS} - V_t \quad (\text{Continuous channel}) \quad (5.4)$$

Either Eq. (5.3) or Eq. (5.4) can be used to ascertain triode-region operation. In words, the n -channel enhancement-type MOSFET operates in the triode region when v_{GS} is greater than V_t and the drain voltage is lower than the gate voltage by at least V_t volts.

In the triode region, the i_D - v_{DS} characteristics can be approximately described by the relationship

$$i_D = K[2(v_{GS} - V_t)v_{DS} - v_{DS}^2] \quad (5.5)$$

in which K is a device parameter given by

$$K = \frac{1}{2} \mu_n C_{OX} \left(\frac{W}{L} \right) \quad (5.6)$$

where μ_n is a physical constant known as the **electron mobility** (its value in this case applies for the electrons in the induced n channel), C_{OX} (called the **oxide capacitance**) is the capacitance per unit area of the gate-to-body capacitor for which the oxide layer serves as dielectric, L is the length of the channel, and W is its width (refer to Fig. 5.1). Since for a given fabrication process the quantity $(\frac{1}{2} \mu_n C_{OX})$ is a constant (approximately $10 \mu\text{A}/\text{V}^2$ for the standard NMOS process with a $0.1\text{-}\mu\text{m}$ oxide thickness), the **aspect ratio** of the device (W/L) determines its conductivity parameter K . Note from Eq. (5.5) that K has the units A/V^2 .

If v_{DS} is sufficiently small so that we can neglect the v_{DS}^2 term in Eq. (5.5), we obtain for the i_D - v_{DS} characteristics near the origin the relationship

$$i_D \approx 2K(v_{GS} - V_t)v_{DS} \quad (5.7)$$

This linear relationship represents the operation of the MOS transistor as a linear resistance r_{DS} ,

$$r_{DS} \equiv \frac{v_{DS}}{i_D} = [2K(v_{GS} - V_t)]^{-1} \quad (5.8)$$

whose value is controlled by v_{GS} . We discussed this region of operation in the previous section (refer to Fig. 5.4).

To operate the MOSFET in the saturation region a channel has to be induced,

$$v_{GS} \geq V_t \quad (\text{Induced channel}) \quad (5.9)$$

and pinched off at the drain end by raising v_{DS} to a value that results in the gate-to-drain voltage falling below V_t ,

$$v_{GD} \leq V_t \quad (\text{Pinched-off channel}) \quad (5.10)$$

This condition can be expressed explicitly in terms of v_{DS} as

$$v_{DS} \geq v_{GS} - V_t \quad (\text{Pinched-off channel}) \quad (5.11)$$

In words, the n -channel enhancement-type MOSFET operates in the saturation region when v_{GS} is greater than V_t and the drain voltage does not fall below the gate voltage by more than V_t volts.

The boundary between the triode region and the saturation region is characterized by

$$v_{DS} = v_{GS} - V_t \quad (\text{Boundary}) \quad (5.12)$$

Substituting this value of v_{DS} into Eq. (5.5) gives the saturation value of the current i_D as

$$i_D = K(v_{GS} - V_t)^2 \quad (5.13)$$

Thus in saturation the MOSFET provides a drain current whose value is independent of the drain voltage v_{DS} and is determined by the gate voltage v_{GS} according to the square-law relationship in Eq. (5.13), a sketch of which is shown in Fig. 5.11. Thus the saturated

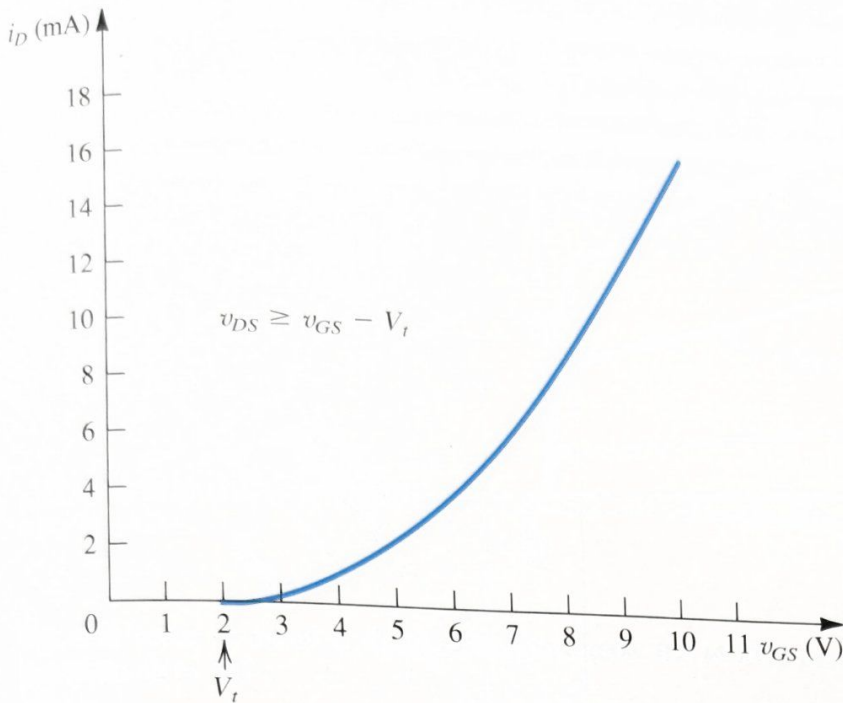


Fig. 5.11 The i_D - v_{GS} characteristic for an enhancement-type NMOS transistor in saturation ($V_t = 2$ V, $K = 0.25$ mA/V²).

MOSFET behaves as an ideal current source whose value is controlled by v_{GS} according to the nonlinear relationship in Eq. (5.13). Figure 5.12 shows a circuit representation of this view of the MOSFET operation in the saturation region. Note that this is a large-signal equivalent circuit model.

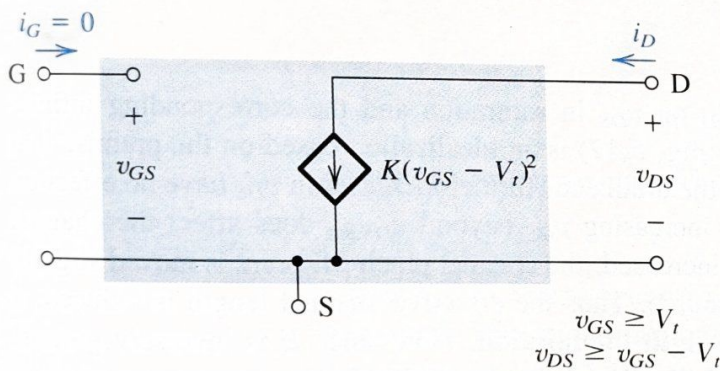


Fig. 5.12 Large-signal equivalent circuit model of an n -channel MOSFET operating in the saturation region.

Referring back to the i_D - v_{DS} characteristics in Fig. 5.10(b), we note that the boundary between the triode and the saturation regions is shown as a broken-line curve. Since this curve is characterized by $v_{DS} = v_{GS} - V_t$, its equation can be found by substituting for $v_{GS} - V_t$ by v_{DS} in either the triode-region equation (Eq. 5.5) or the saturation-region equation (Eq. 5.13). The result is

$$i_D = K v_{DS}^2 \quad (5.14)$$

It should be noted that the characteristics depicted in Figs. 5.10, 5.11, and 5.4 are for a MOSFET with $K = 0.25 \text{ mA/V}^2$ and $V_t = 2 \text{ V}$.

Finally, the chart in Fig. 5.13(a) shows the relative levels that the terminal voltages of the enhancement-type NMOS transistor must have for operation in the triode region and in the saturation region. The same information is presented in a different way in Fig. 5.13(b).

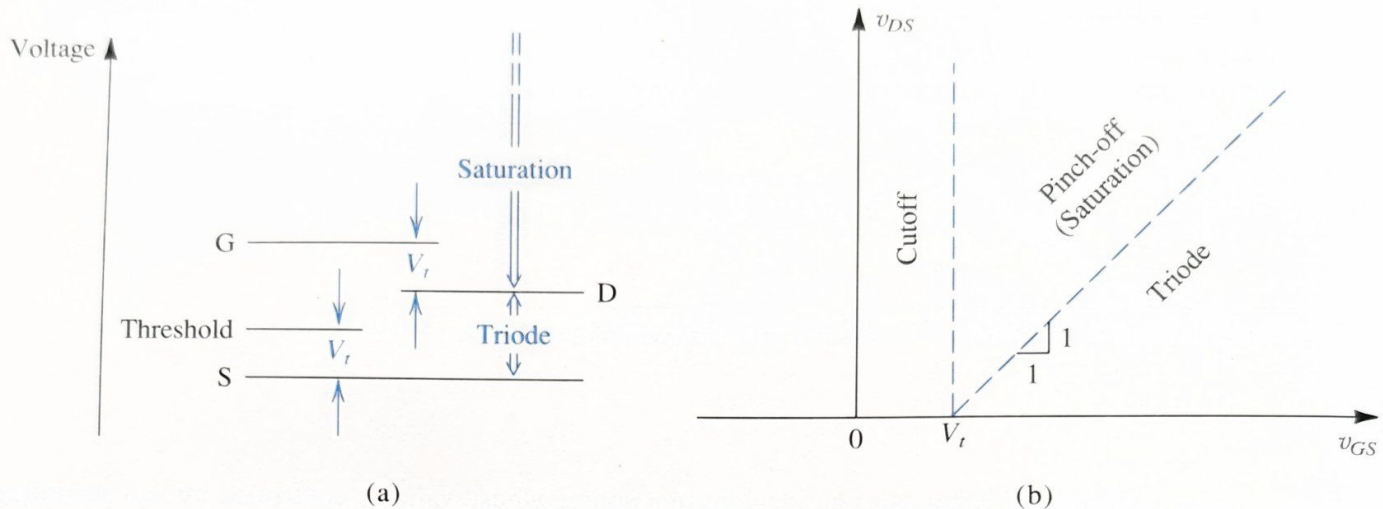


Fig. 5.13 (a) The relative levels of the terminal voltages of the enhancement NMOS transistor for operation in the triode region and in the saturation region. **(b)** Illustrating the different regions of operation on the v_{DS} – v_{GS} plane.

Finite Output Resistance in Saturation

The complete independence of i_D on v_{DS} in saturation and the corresponding infinite output resistance at the drain (see Fig. 5.12) is an idealization based on the premise that once the channel is pinched off at the drain end, further increases in v_{DS} have no effect on the channel's shape. In practice, increasing v_{DS} beyond $v_{DS,sat}$ does affect the channel somewhat. Specifically, as v_{DS} is increased, the channel pinch-off point is moved slightly away from the drain toward the source. Thus the effective channel length is reduced, a phenomenon known as **channel-length modulation**. Now since K is inversely proportional to the channel length (Eq. 5.6), K and, correspondingly, i_D , increases with v_{DS} .

A typical set of i_D – v_{DS} characteristics showing the effect of channel length modulation is displayed in Fig. 5.14. The slight linear dependence of i_D on v_{DS} in the saturation region can be analytically accounted for by incorporating the factor $1 + \lambda v_{DS}$ in the i_D equation as follows:

$$i_D = K(v_{GS} - V_t)^2(1 + \lambda v_{DS}) \quad (5.15)$$

where the positive constant λ is a MOSFET parameter. From Fig. 5.14 we observe that the straight-line i_D – v_{DS} characteristics in saturation, when extrapolated, intercept the v_{DS} -axis at the point $v_{DS} = -1/\lambda \equiv -V_A$, where V_A is a positive voltage similar to the

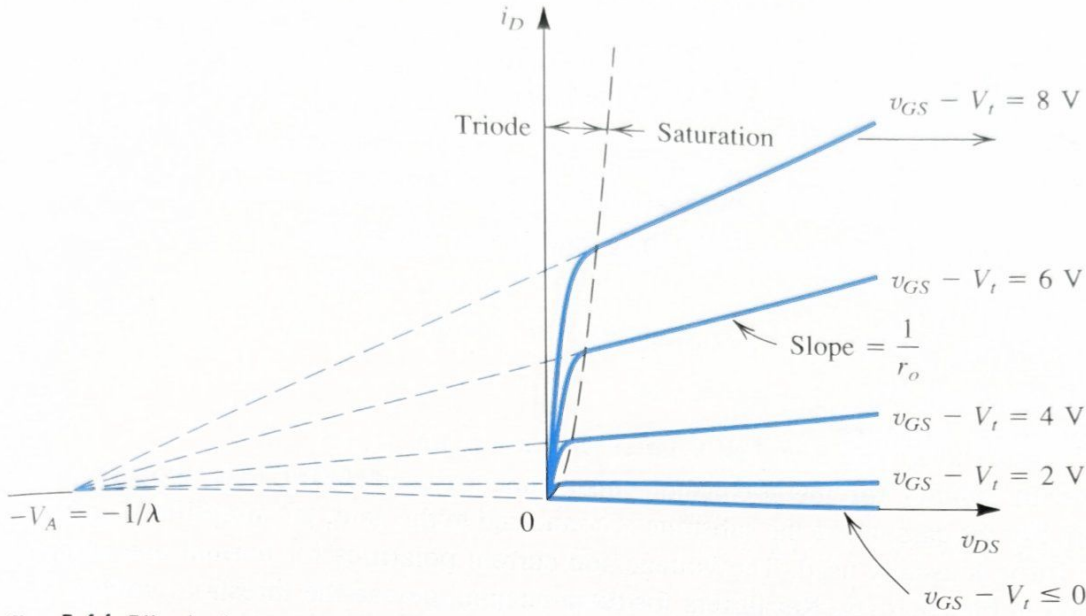


Fig. 5.14 Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A is typically in the range 30 to 200 V.

Early voltage in a BJT. Typically $\lambda = 0.005$ to 0.03 V^{-1} , and, correspondingly, V_A is in the range 200 to 30 volts.

It should be obvious that channel-length modulation makes the output resistance in saturation finite. Defining the output resistance r_o as⁶

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}} \right]^{-1}_{v_{GS}=\text{constant}} \quad (5.16)$$

results in

$$r_o = [\lambda K (V_{GS} - V_t)^2]^{-1}$$

which can be approximated by

$$r_o \approx [\lambda I_D]^{-1} \quad (5.17)$$

where I_D is the current corresponding to the particular value of v_{GS} for which r_o is being evaluated. The approximation in Eq. (5.17) is based on neglecting the effect of the factor $1 + \lambda v_{DS}$ in Eq. (5.15) on the value of r_o —a second-order approximation. Equation (5.17) can be written in the alternative form

$$r_o \approx V_A / I_D \quad (5.18)$$

Thus the output resistance is inversely proportional to the dc bias current I_D . Finally, we show in Fig. 5.15 the large-signal equivalent circuit model incorporating r_o .

⁶In this book we use r_o to represent the incremental output resistance in saturation and r_{DS} to represent the drain-to-source resistance in the triode region for small v_{DS} .

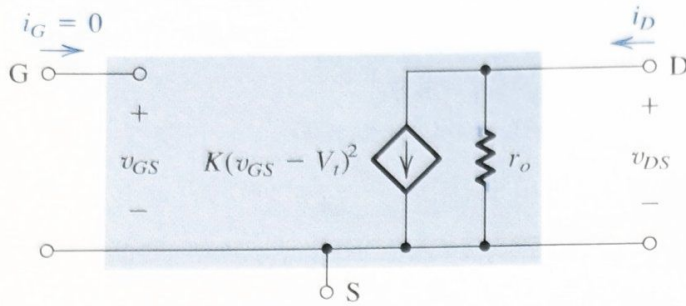


Fig. 5.15 Large-signal equivalent circuit model of the n -channel MOSFET in saturation, incorporating the output resistance r_o . The output resistance models the linear dependence of i_D on v_{DS} and is given by $r_o = V_A/[K(V_{GS} - V_t)^2] \approx V_A/I_D$.

Characteristics of the p -Channel MOSFET

The circuit symbol for the p -channel enhancement-type MOSFET is shown in Fig. 5.16(a). For the case where the substrate is connected to the gate, the simplified symbol of Fig. 5.16(b) is usually used. The voltage and current polarities for normal operation are indicated in Fig. 5.16(c). Recall that for the p -channel device the threshold voltage V_t is negative. To induce a channel we apply a gate voltage that is more negative than V_t ,

$$v_{GS} \leq V_t \quad (\text{Induced channel}) \quad (5.19)$$

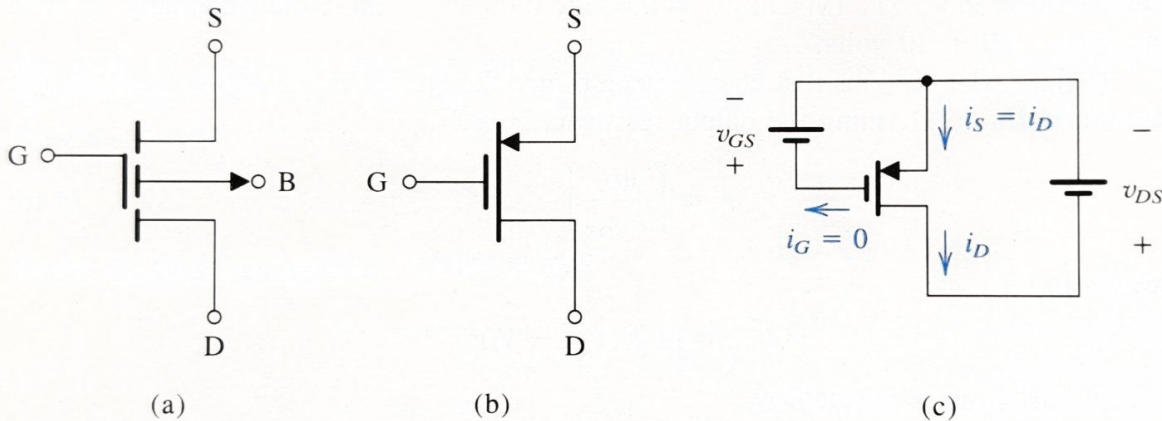


Fig. 5.16 (a) Circuit symbol for the p -channel enhancement-type MOSFET. (b) Simplified circuit symbol for the case where the substrate (body, B) is connected to the source. (c) The MOSFET with voltages applied and the directions of current flow indicated. Note that v_{GS} and v_{DS} are negative and i_D flows out of the drain terminal.

and apply a drain voltage that is more negative than the source voltage (i.e., v_{DS} is negative or, equivalently, v_{SD} is positive). The current i_D flows out of the drain terminal, as indicated in the figure. To operate in the triode region v_{DS} must satisfy

$$v_{DS} \geq v_{GS} - V_t \quad (\text{Continuous channel}) \quad (5.20)$$

that is, the drain voltage must be higher than the gate voltage by at least $|V_t|$. The current i_D is given by the same equation as for NMOS, Eq. (5.5),

$$i_D = K[2(v_{GS} - V_t)v_{DS} - v_{DS}^2] \quad (5.21)$$

where v_{GS} , V_t , and v_{DS} are negative and K is given by

$$K = \frac{1}{2} \mu_p C_{OX} \left(\frac{W}{L} \right) \quad (5.22)$$

where μ_p is the mobility of holes in the induced p channel. Typically, $\mu_p \approx \frac{1}{2} \mu_n$, with the result that for the same W/L ratio a PMOS transistor has half the value of K as the NMOS device.

To operate in saturation, v_{DS} must satisfy

$$v_{DS} \leq v_{GS} - V_t \quad (\text{Pinched-off channel}) \quad (5.23)$$

that is, the drain voltage must be lower than (gate voltage $+|V_t|$). The current i_D is given by the same equation used for NMOS, Eq. (5.15),

$$i_D = K(v_{GS} - V_t)^2(1 + \lambda v_{DS}) \quad (5.24)$$

where v_{GS} , V_t , λ , and v_{DS} are all negative.

Finally, the chart in Fig. 5.17 shows the relative levels that the terminal voltages of the enhancement-type PMOS transistor must have for operation in the triode region and in the saturation region.

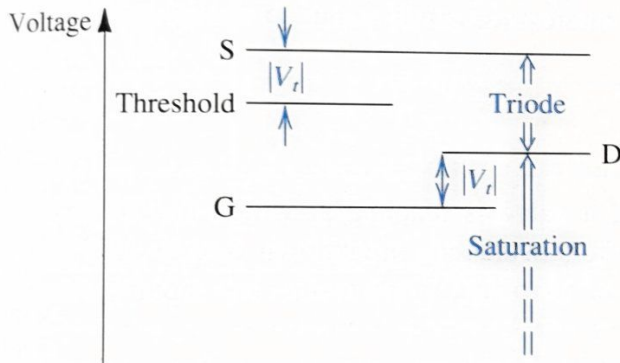


Fig. 5.17 The relative levels of the terminal voltages of the enhancement-type PMOS transistor for operation in the triode region and in the saturation region.

The Role of the Substrate—The Body Effect

In many applications the substrate (or body) terminal B is connected to the source terminal, which results in the pn junction between the substrate and the induced channel (see Fig. 5.5) having a constant reverse bias. In such a case the substrate does not play any role in circuit operation and its existence can be ignored altogether.

In integrated circuits, however, the substrate is usually common to many MOS transistors. In order to maintain the reverse-bias condition on the substrate-to-channel junction, the substrate is usually connected to the most negative power supply in an NMOS circuit (the most positive in a PMOS circuit). The resulting reverse-bias voltage between source and body (V_{SB} in an n -channel device) will have an effect on device operation. To appreciate this fact consider an NMOS transistor and let its substrate be made negative relative to the source. The reverse bias voltage will widen the depletion region (refer to Fig. 5.2). This in turn reduces the channel depth. To return the channel to its former state, v_{GS} has to be increased.

The effect of V_{SB} on the channel can be most conveniently represented as a change in the threshold voltage V_t . Specifically, it can be shown [see Tsividis (1987)] that increasing the reverse substrate bias voltage V_{SB} results in an increase in V_t according to the relationship

$$V_t = V_{t0} + \gamma \left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f} \right] \quad (5.25)$$

where V_{t0} is the threshold voltage for $V_{SB} = 0$; γ is a process parameter, typically $0.5 \text{ V}^{1/2}$; ϕ_f is a physical parameter with $(2\phi_f)$ typically 0.6 V .

Equation (5.25) indicates that an incremental change in V_{SB} gives rise to an incremental change in V_t , which in turn results in an incremental change in i_D even though v_{GS} might have been kept constant. It follows that the body voltage controls i_D ; thus the body acts as another gate for the MOSFET, a phenomenon known as the **body effect**. The body effect can cause considerable degradation in circuit performance, as will be shown in Section 5.9.

Temperature Effects

Both V_t and K are temperature sensitive. The magnitude of V_t decreases by about 2 mV for every 1°C rise in temperature. This decrease in $|V_t|$ gives rise to a corresponding increase in drain current as temperature is increased. However, because K decreases with temperature and its effect is a dominant one, the overall observed effect of a temperature increase is a *decrease* in drain current. This very interesting result is put to use in applying the MOSFET in power circuits (Chapter 9).

Breakdown and Input Protection

As the voltage on the drain is increased, a value is reached at which the *pn* junction between the drain region and substrate suffers avalanche breakdown (see Section 3.11). This breakdown usually occurs at voltages of 50 to 100 V and results in a rapid increase in current.

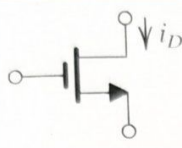
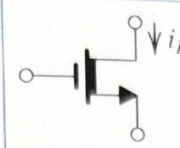
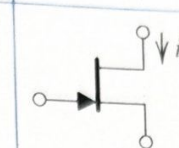
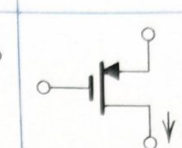
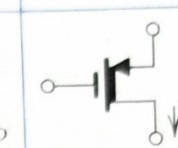
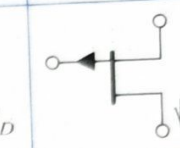
Another breakdown effect that occurs at lower voltages (about 20 V) in modern devices is called **punch-through**. It occurs in devices with relatively short channels when the drain voltage is increased to the point that the depletion region surrounding the drain region extends through the channel to the source. The drain current then increases rapidly. Punch-through does not normally result in permanent damage to the device.

Yet another kind of breakdown occurs when the gate-to-source voltage exceeds about 50 V . This is the breakdown of the gate oxide and results in permanent damage to the device. Although 50 V is high, it must be remembered that the MOSFET has a very high input impedance, and thus small amounts of static charge accumulating on the gate capacitor can cause this breakdown voltage to be exceeded.

To prevent the accumulation of static charge on the gate capacitor of a MOSFET, gate protection devices are usually included at the input of MOS integrated circuits. The protection mechanism invariably makes use of clamping diodes (see Fig. 13.32).

For future reference and to permit comparison of the various types of FETs studied in this chapter, Table 5.1 provides a summary of the important formulas and voltage polarities. (Depletion MOSFETs and JFETs will be studied shortly.)

Table 5.1 SUMMARY OF THE CHARACTERISTICS OF FIELD-EFFECT TRANSISTORS

	n-Channel			p-Channel		
	Enhancement MOSFET	Depletion MOSFET	JFET	Enhancement MOSFET	Depletion MOSFET	JFET
Circuit Symbol						
V_t	+	-	-	-	+	+
K	$\frac{1}{2} \mu_n C_{OX}(W/L)$		I_{DSS}/V_P^2	$\frac{1}{2} \mu_p C_{OX}(W/L)$		I_{DSS}/V_P^2
To turn transistor on	$v_{GS} > V_t$			$v_{GS} < V_t$		
v_{DS}	+			-		
To operate in the triode region	$v_{DS} \leq v_{GS} - V_t$			$v_{DS} \geq v_{GS} - V_t$		
To operate in the saturation region	$v_{DS} \geq v_{GS} - V_t$			$v_{DS} \leq v_{GS} - V_t$		
$\lambda = 1/V_A$	+			-		
In triode region	$i_D = K[2(v_{GS} - V_t)v_{DS} - v_{DS}^2]$					
In saturation region	$i_D = K(v_{GS} - V_t)^2(1 + \lambda v_{DS})$					
r_o	$ V_A /I_D$					

Exercises 5.1 An enhancement-type NMOS transistor with $V_t = 2$ V has its source grounded and a 3-V dc source connected to the gate. In what region of operation does the device operate for (a) $V_D = +0.5$ V? (b) $V_D = 1$ V? (c) $V_D = 5$ V?

Ans. (a) Triode; (b) Saturation; (c) Saturation

5.2 If the NMOS device in Exercise 5.1 has $\mu_n C_{OX} = 20 \mu\text{A}/\text{V}^2$, $W = 100 \mu\text{m}$, and $L = 10 \mu\text{m}$, find the value of K . Also find the value of drain current that results in each of the three cases (a), (b), and (c) specified in Exercise 5.1. Neglect the dependence of i_D on v_{DS} in saturation.

Ans. $0.1 \text{ mA}/\text{V}^2$; (a) $75 \mu\text{A}$; (b) $100 \mu\text{A}$; (c) $100 \mu\text{A}$

5.3 An enhancement-type NMOS transistor with $V_t = 2$ V conducts a current $i_D = 1$ mA when $v_{GS} = v_{DS} = 3$ V. Neglecting the dependence of i_D on v_{DS} in saturation, find the value of i_D for $v_{GS} = 4$ V and $v_{DS} = 5$ V. Also, calculate the value of the drain-to-source resistance r_{DS} for small v_{DS} and $v_{GS} = 4$ V.

Ans. 4 mA ; 250Ω

5.4 An enhancement-type MOSFET with $K = 0.1 \text{ mA/V}^2$, $V_t = 1.5 \text{ V}$, and $\lambda = 0.02 \text{ V}^{-1}$ is operated at $v_{GS} = 3.5 \text{ V}$. Find the drain current obtained at $v_{DS} = 2 \text{ V}$ and at $v_{DS} = 10 \text{ V}$. Determine the output resistance r_o at this value of v_{GS} .

Ans. 0.416 mA; 0.480 mA; 125 k Ω

5.5 An IC NMOS transistor has $W = 100 \text{ }\mu\text{m}$, $L = 10 \text{ }\mu\text{m}$, $\mu_n C_{OX} = 20 \text{ }\mu\text{A/V}^2$, $V_A = 100 \text{ V}$, $\gamma = \frac{1}{2} \text{ V}^{1/2}$, $V_{t0} = 1 \text{ V}$, and $2\phi_f = 0.6 \text{ V}$. Calculate the values of K and of V_t at $V_{SB} = 4 \text{ V}$. Also, for $V_{GS} = 3 \text{ V}$ and $V_{DS} = 5 \text{ V}$, calculate I_D for $V_{SB} = 0 \text{ V}$ and for $V_{SB} = 4 \text{ V}$. What is the output resistance r_o for each of the two cases?

Ans. 0.1 mA/V²; 1.7 V; 0.420 mA; 0.177 mA; 250 k Ω ; 592 k Ω

5.6 An enhancement-type PMOS transistor has $K = 50 \text{ }\mu\text{A/V}^2$ and $V_t = -2 \text{ V}$. If the gate is grounded and the source is connected to +5 V, what is the highest voltage that can be applied to the drain while the device operates in saturation? Neglecting the finite r_o , find the drain current for $v_D = -5 \text{ V}$.

Ans. +2 V; 0.45 mA

5.3 THE DEPLETION-TYPE MOSFET

In this section we briefly discuss another type of MOSFET, the depletion-type MOSFET. Its structure is similar to that of the enhancement-type MOSFET with one important difference: The depletion MOSFET has a physically implanted channel. Thus an n -channel depletion type MOSFET has an n -type silicon region connecting the n^+ source and the n^+ drain regions at the top of the p -type substrate. Thus if a voltage v_{DS} is applied between drain and source, a current i_D flows for $v_{GS} = 0$. In other words, there is no need to induce a channel, unlike the case of the enhancement MOSFET.

The channel depth and hence its conductivity can be controlled by v_{GS} in exactly the same manner as in the enhancement-type device. Applying a positive v_{GS} enhances the channel by attracting more electrons into it. Here, however, we also can apply a negative v_{GS} , which causes electrons to be repelled from the channel; and thus the channel becomes shallower and its conductivity decreases. The negative v_{GS} is said to **deplete** the channel of its charge carriers, and this mode of operation (negative v_{GS}) is called **depletion mode**. As the magnitude of v_{GS} is increased in the negative direction, a value is reached at which the channel is completely depleted of charge carriers and i_D is reduced to zero even though v_{DS} may be still applied. This negative value of v_{GS} is the threshold voltage of the n -channel depletion-type MOSFET.

The description above suggests (correctly) that a depletion-type MOSFET can be operated in the enhancement mode by applying a positive v_{GS} and in the depletion mode by applying a negative v_{GS} . The i_D - v_{DS} characteristics are similar to those for the enhancement device except that V_t of the n -channel depletion device is negative.

Figure 5.18(a) shows the circuit symbol for the n -channel depletion-type MOSFET. This symbol differs from that of the enhancement-type device in only one respect: The vertical line representing the channel is solid, signifying that a physical channel exists. When the body (B) is connected to the source (S) the simplified symbol shown in Fig. 5.18(b) can be used. This symbol differs from the corresponding one for the enhancement device in that a shaded area is included to denote the implanted channel.

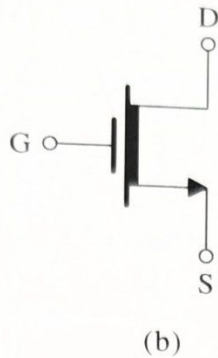
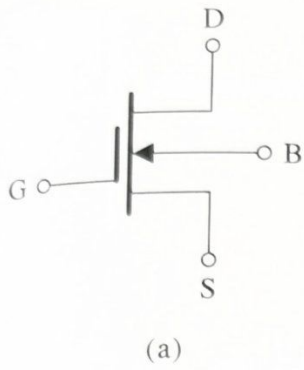


Fig. 5.18 (a) Circuit symbol for the n -channel depletion-type MOSFET. (b) Simplified circuit symbol applicable for the case the substrate (B) is connected to the source (S).

The i_D - v_{DS} characteristics of a depletion-type n -channel MOSFET for which $V_t = -4$ V and $K = 1$ mA/V² are sketched in Fig. 5.19(b). Although these characteristics do not show the dependence of i_D on v_{DS} in saturation, such dependence exists and is identical to the case for the enhancement-type device. Observe that because the threshold voltage V_t is negative, the depletion NMOS will operate in the triode region as long as the drain voltage does not exceed the gate voltage by more than $|V_t|$. For it to operate in saturation, the drain voltage must be greater than the gate voltage by at least $|V_t|$ volts. The chart in Fig. 5.20(a) shows the relative levels of the terminal voltages of the depletion NMOS transistor for the two regions of operation. Another way of representing this information is given in Fig. 5.20(b).

Figure 5.19(c) shows the i_D - v_{GS} characteristics in saturation, indicating both the depletion and enhancement modes of operation.

The current-voltage characteristics of the depletion-type MOSFET are described by the equations given in the previous section for the enhancement device except that, for an n -channel depletion device, V_t is negative.

Another parameter for the depletion MOSFET is the value of drain current obtained in saturation with $v_{GS} = 0$. It is denoted I_{DSS} and is indicated in Fig. 5.19(b) and (c). It can be shown that

$$I_{DSS} = K V_t^2 \quad (5.26)$$

Depletion-type MOSFETs can be fabricated on the same integrated-circuit chip as enhancement-type devices, resulting in circuits with improved characteristics, as will be shown in a later section.

In the above we discussed only n -channel depletion devices. Depletion PMOS transistors are available in discrete form and operate in a manner similar to their n -channel counterparts except that the polarities of all voltages (including V_t) are reversed. Also, in a p -channel device, i_D flows from source to drain, entering the source terminal and leaving by way of the drain terminal. As a summary, we show in Fig. 5.21 sketches of the i_D - v_{GS} characteristics of enhancement and depletion MOSFETs of both polarities (operating in saturation). Finally, we draw the reader's attention to the summary of the characteristics of depletion MOSFETs, presented in Table 5.1.

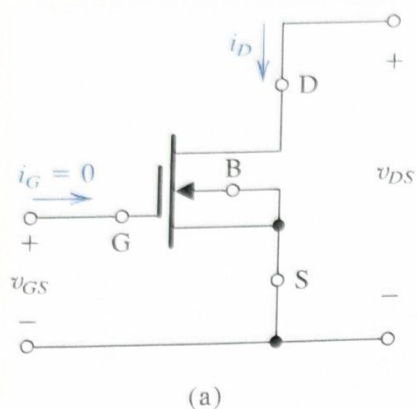
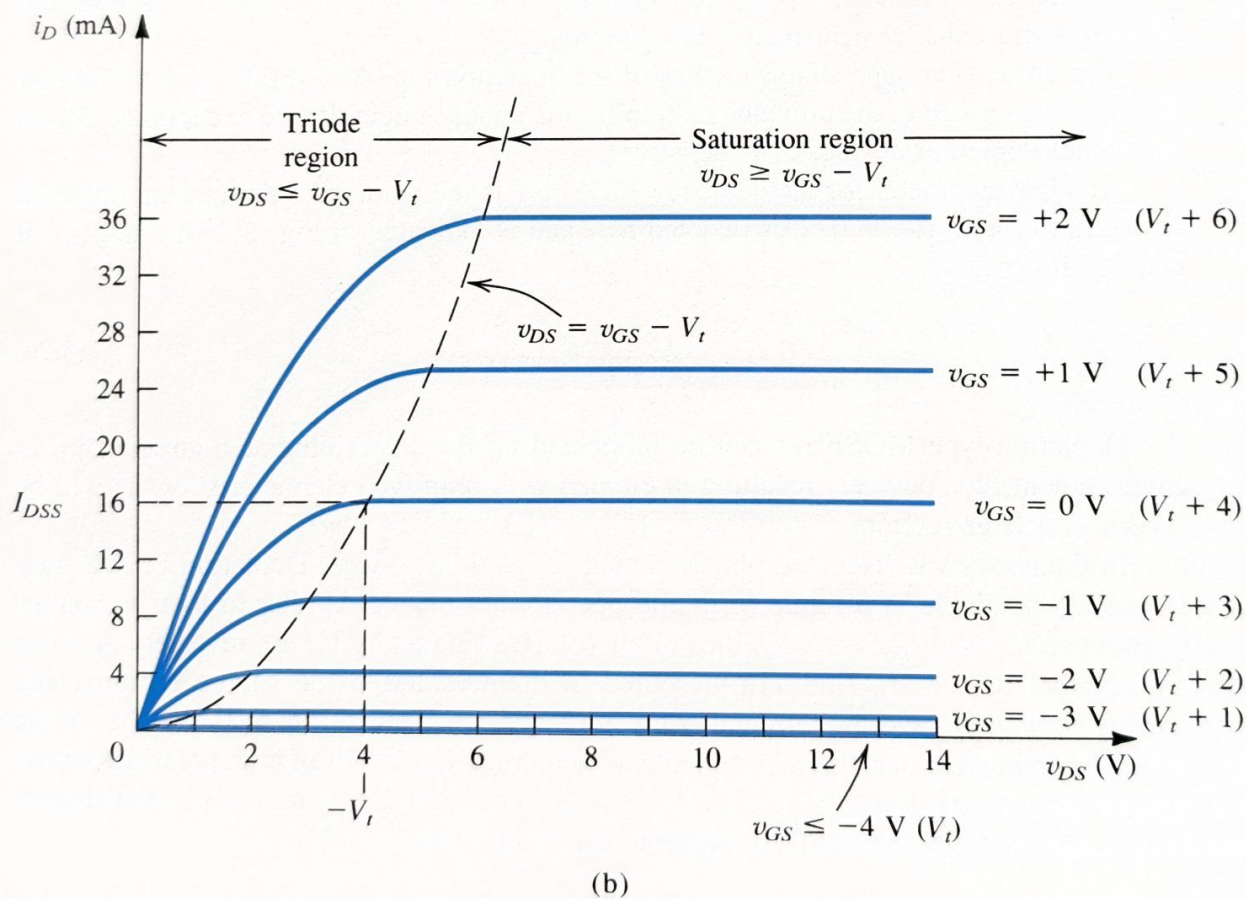
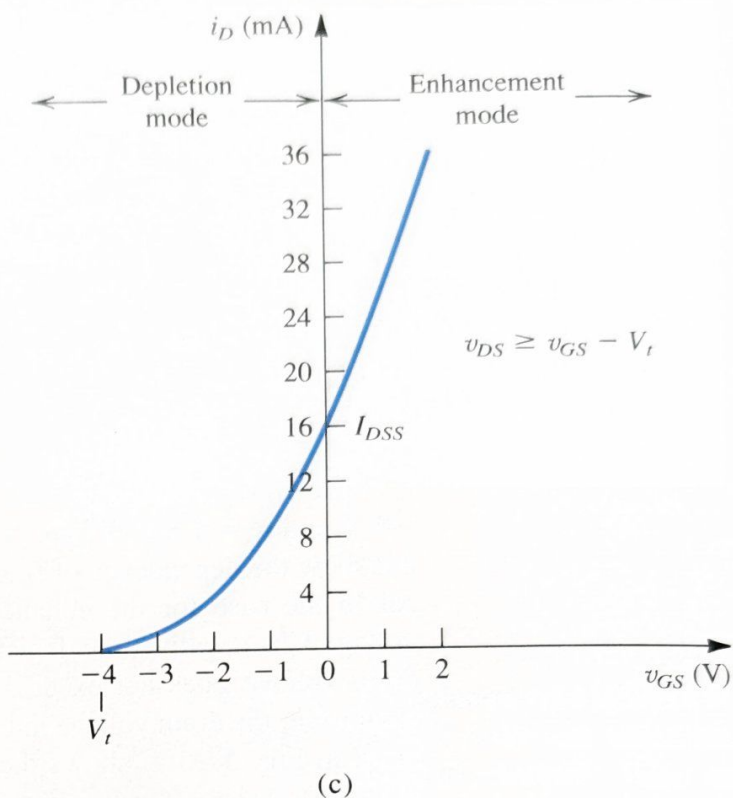


Fig. 5.19 The current-voltage characteristics of a depletion-type n -channel MOSFET for which $V_t = -4\text{ V}$ and $K = 1\text{ mA/V}^2$: (a) transistor with current and voltage polarities indicated; (b) the i_D - v_{DS} characteristics; (c) the i_D - v_{GS} characteristics in saturation.



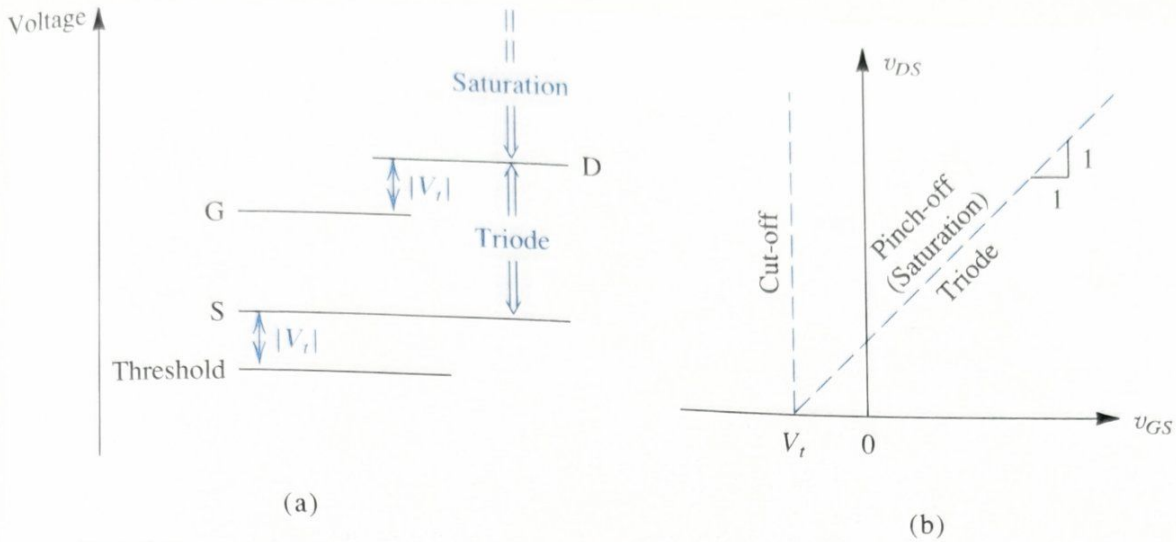


Fig. 5.20 (a) The relative levels of terminal voltages of a depletion-type NMOS transistor for operation in the triode and the saturation regions. The case shown is for operation in the enhancement mode (v_{GS} is positive). (b) Representation of the regions of operation on the v_{DS} - v_{GS} plane.

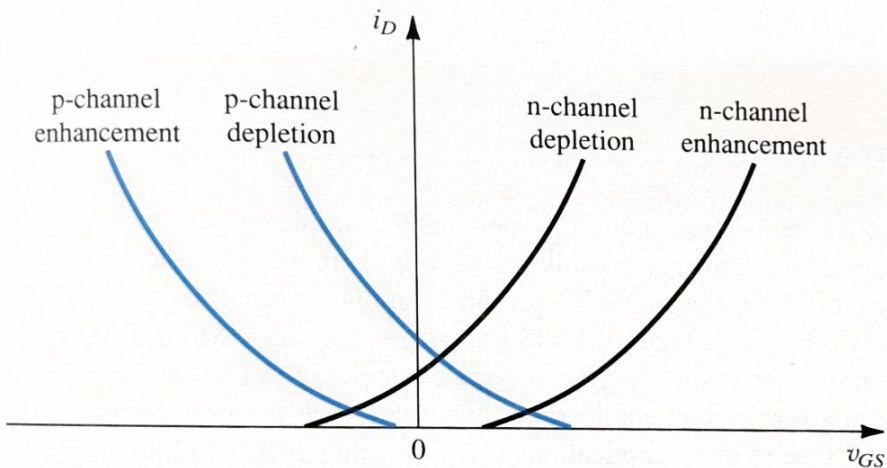


Fig. 5.21 Sketches of the i_D - v_{GS} characteristics for MOSFETs of enhancement and depletion types, of both polarities (operating in saturation). Note that the characteristic curves intersect the v_{GS} axis at V_t . Also note that somewhat different values of $|V_t|$ are shown for n -channel and p -channel devices.

Exercises 5.7 For a depletion-type NMOS transistor with $V_t = -2$ V and $K = 2$ mA/V², find the minimum v_{DS} required to operate in the saturation region when $v_{GS} = +1$ V. What is the corresponding value of i_D ?

Ans. 3 V; 18 mA

5.8 The depletion-type MOSFET in Fig. E5.8 has $K = 2$ mA/V² and $V_t = -2$ V. Neglecting the effect of v_{DS} on i_D in the saturation region, find the voltage that will appear at the source terminal.